Social and Professional Issues in Computing, Course Code - CSE326

CASE STUDY REPORT

ON

Enterprise IT Infrastructure & Security Architecture At Star Tech & Engineering Ltd (STEL)

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INTERVIEWEE DETAILS

Mr. Ariful Islam
Manager, Presales (Network & Security)
Star Tech & Engineering Ltd (STEL)

This Case Study Report is Submitted as a Course Assignment for CSE326: Social and Professional Issues in Computing

Submitted To

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A Case Study on Enterprise IT Infrastructure, Security Architecture, and Professional Ethics

An Academic Industry Visit Report on Star Tech & Engineering Ltd (STEL)

1. Introduction

Exposure to industrial applications of theories related to computing is the backbone of learning practical applications of theoretical computing ideas. For the academic course on CSE326: Social and Professional Issues in Computing at Daffodil International University, we, the students, went through a formal industrial visit to Star Tech & Engineering Ltd (STEL). The main purpose of the visit was to explore the ways in which IT companies manage their network systems along with ethical and legal issues in relation to privacy laws, reliability and intellectual property rights. This case study is an analysis of what we learned from our formal interview with **Mr. Ariful Islam, the Manager of Presales (Network & Security) of STEL.**

2. About the company

Star Tech & Engineering Ltd (STEL) is one of the most reputable technology firms and system integrators of Bangladesh. Although popular in the B2C technology sector, its Enterprise Division focuses on designing and implementing IT environments for B2B enterprise networking and cybersecurity architecture. Headquartered in Dhaka, the Enterprise Division of STEL integrates large scale system infrastructures for financial institutions, conglomerate firms, and government organizations through high-end hardware and security products supplied by renowned vendors like Cisco, Fortinet, Palo Alto Networks, F5, and Radware.

3. Observation & Technical Analysis

In the course of the field visit, our team identified the processes and architectures used by STEL's Presales and Security Engineering departments. The technical analysis of the processes used by the company is analyzed in key academic evaluation categories as follows:

A. Privacy and Data Collection Principles

- **Zero Trust Network Access (ZTNA):** In implementing ZTNA, STEL follows the principle of "Zero Tolerance." Network access will not be based on employee identification alone. It will be based on the ongoing process of employee identity validation and health check via Network Access Control (NAC) and Two-Factor Authentication (2FA).
- **Device Posture Validation:** Prior to allowing network access to company applications, the device posture of the end-point is first validated in terms of the condition of its Operating System (OS). If the end-point device is not up-to-date with security patches, automatic network entry is prohibited.
- **Data Minimization:** STEL strictly observes data minimization during project scoping and network traffic. The engineering team gets more high-level metadata (ISPs, bandwidth

utilization, number of simultaneous users) but does not probe further to the extent of getting actual user data or even database itself.

B. Network Security and Threat Mitigation

- **DDoS Mitigation Architecture:** In response to the issue of volumetric DDoS attacks, the STEL framework uses enterprise-grade hardware (such as Radware, F5, Fortinet) to detect these threats. Through the use of AI-based techniques for detecting patterns of traffic that deviate from the norm, any malicious traffic is blocked at the perimeter to avoid exhaustion of the client's ISP bandwidth pipe.
- **Encrypted Remote Access (SSL VPN):** In order to protect the company's remote operations from MitM packet sniffing, STEL uses SSL VPN solutions for enterprises (such as Cisco AnyConnect, Palo Alto GlobalProtect, FortiClient), where all information being transferred across the internet is encrypted and cannot be understood without the decryption key.

C. System Fails, QA and Reliability

- **High Availability (HA) and Hardware Redundancy:** In order to avoid single points of failure for mission critical network infrastructure, core hardware (routers, firewalls, and switches) is implemented in Active-Standby HA pairs. Traffic failover to the second device happens automatically in case of failure of the primary device due to hardware problems without any disruption of the service.
- **Disaster Recovery (DR) Procedures:** In cases of major failure at the entire site level, STEL proposes that there be physically separated Data Center (DC) and Disaster Recovery (DR) locations. Ensuring that the DC and DR locations be physically apart will provide business continuity during regional disasters.
- **Migration Quality Assurance (QA):** Before performing the actual migration of the network, STEL constructs a testing/staging environment. Tests for the flow of traffic will be performed with isolated endpoints. Cutover from old to new systems will take place only after ensuring 100% operability in the testing/staging environment.

D. Intellectual Property (IP) and Licensing

- **True Software Licensing:** STEL is an authorized channel partner of leading Original Equipment Manufacturers (OEMs) worldwide. In order to protect software copyright and patents, licenses and subscriptions are provided to the end-customer through their vendor portals. This ensures there is no interference from intermediaries and any possibility of unlicensed software is ruled out.
- **Hazards of Open Source Software Crackers:** STEL recommends against the use of unlicensed open source software cracks for enterprise clients.

4. Professional and Ethical Issues

The use of infrastructure requires the IT professional to be responsible for several ethical aspects:

- **Confidentiality of Clients and Non-Disclosure Agreement:** Integration experts often have access to sensitive diagrams of the network. STEL adheres to strict non-disclosure policy guidelines. Network engineers perform troubleshooting through remote-desktop sessions under monitoring without gaining root credentials permanently.

- **Ethical Presales Risk Evaluation:** It is the responsibility of the engineer to inform clients about the risks of using open-source software versus paid enterprise solutions so that clients can make informed decisions.

5. Social Effects and Future Technologies

- **Effect of Artificial Intelligence (AI) on Cyber Security:** The effect of using AI speeds up the detection of the threat, however, at the same time, it gives power to hackers. Hackers use high-performance GPUs for decrypting algorithms on traditional VPN tunnels.
- **Preparing for Future Technologies:** In order to deal with quantum-computing and AI decryption attacks, the industry is shifting towards Post-Quantum Cryptography (PQC).

6. Recommendations

Considering the above case studies, the following recommendations are presented:

- **PQC Implementation:** Enterprise clients should start planning for implementing PQC in order to protect their confidential data from AI decryption in the long run.
- **Endpoint Posture Assessment:** Companies that operate with remote work models should require conducting periodic automated assessments of the endpoint postures before enabling SSL VPN access.
- **Ethics Courses in Academics:** Hands-on courses about NDA and remote work ethics should be included into the academic curriculum along with laboratory work on networks.

7. Conclusion

Our industry visit to Star Tech & Engineering Ltd gave us valuable knowledge regarding practical applications of theoretical computing concepts. Through our visit to observe the implementation of Zero Trust architecture, HA redundancy, SSL VPN encryption and licensing protocols, we were able to understand the practical aspects of IT infrastructure operations of any business enterprise.

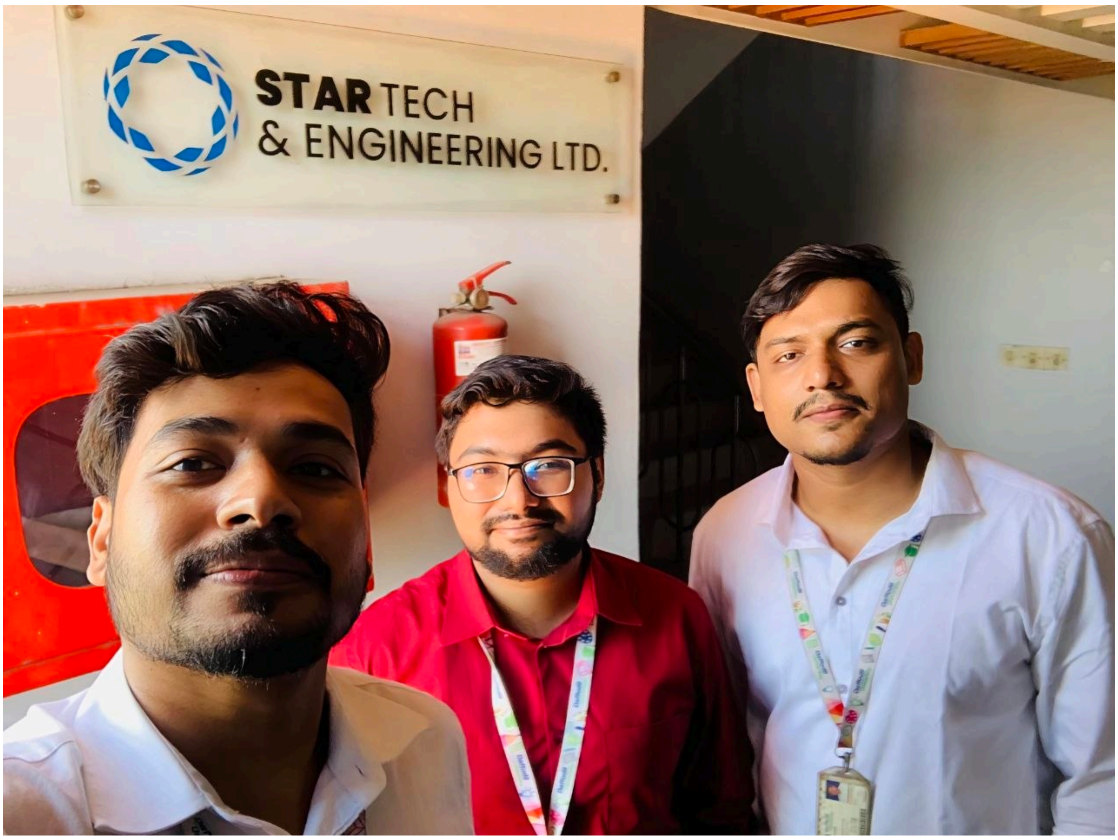
STAR TECH
& ENGINEERING LTD.

Figure: Glimpses of Academic Industry Visit and Interview Session at Star Tech & Engineering Ltd